

ROST, Norway

WMO: 011070

Lat: 67.527N

Lon: 12.104E

Elev: 13

StdP: 14.69

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	23.5	25.7	13.1	10.7	26.4	15.3	12.0	28.1	48.6	35.4	43.6	36.7	15.3	90	0.882

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	5.9	62.8	57.8	60.3	56.3	58.5	55.1	58.4	61.8	56.9	59.7	55.5	57.9	10.5	90

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
56.7	68.5	60.0	55.3	65.0	58.6	53.9	61.9	57.3	25.4	61.8	24.4	60.1	23.5	58.1	64.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
38.8	34.5	30.9	22.5	66.1	3.0	4.3	20.4	69.2	18.7	71.7	17.0	74.2	14.9	77.3		
			20.0	60.1	2.9	2.7	17.9	62.1	16.2	63.6	14.6	65.2	12.5	67.1		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	42.8	33.9	34.6	34.8	39.1	44.0	47.9	53.2	53.7	50.6	44.6	39.8	36.5
	DBStd	8.10	4.54	4.64	4.20	3.89	3.94	3.40	3.74	3.24	3.52	4.15	4.65	4.37
	HDD50	2943	498	431	471	326	191	81	8	4	34	175	306	418
	HDD65	8111	963	851	936	776	651	512	366	352	431	634	756	883
	CDD50	307	0	0	0	0	5	19	106	118	53	6	0	0
	CDD65	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDH74	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDH80	0	0	0	0	0	0	0	0	0	0	0	0	0
Wind	WSAvg	15.6	18.1	18.0	17.8	14.7	14.0	14.1	12.6	12.4	13.6	17.0	16.9	18.4
Precipitation	PrecAvg	35.8	4.3	3.3	3.0	2.0	1.5	1.7	2.1	2.6	3.2	3.8	3.9	4.2
	PrecMax	49.7	7.6	6.1	5.2	3.9	2.5	2.6	3.3	4.3	4.8	7.2	7.8	8.2
	PrecMin	21.1	2.0	1.1	0.5	0.9	0.7	0.6	0.9	1.1	1.2	1.0	0.7	1.1
	PrecStd	6.5	1.5	1.4	1.3	0.7	0.4	0.6	0.7	0.8	0.9	1.7	1.5	1.7
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	45.1	44.8	44.9	50.2	58.5	60.9	67.8	65.5	60.2	55.6	51.1	46.7
		MCWB	43.3	43.0	42.3	44.9	51.8	55.2	60.8	59.9	55.3	50.5	49.2	45.0
	2%	DB	43.8	43.7	43.2	47.6	53.9	56.4	64.2	61.9	58.1	53.6	49.4	45.0
		MCWB	41.7	41.9	40.9	43.6	49.1	52.2	58.6	57.5	54.3	49.9	46.9	42.9
	5%	DB	42.3	42.6	42.1	45.7	51.0	54.2	61.3	59.9	56.7	51.9	47.7	43.7
		MCWB	40.3	40.7	39.9	42.7	47.2	50.7	57.1	56.3	53.4	48.7	44.9	41.2
	10%	DB	40.7	41.3	40.9	44.4	48.9	52.6	58.8	58.5	55.4	50.4	46.1	42.4
		MCWB	38.4	39.4	38.7	41.8	45.7	49.7	55.4	55.4	52.7	47.5	43.3	39.6
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	43.9	43.7	43.0	45.6	52.1	55.4	61.8	60.5	56.7	51.9	49.9	45.6
		MCDB	44.7	44.3	44.1	48.8	57.5	59.6	66.1	64.2	58.6	54.4	51.5	45.8
	2%	WB	42.1	42.3	41.5	44.2	49.4	52.8	59.0	58.3	54.9	50.3	47.5	43.7
		MCDB	43.4	43.4	42.7	46.3	53.3	55.6	63.0	61.0	56.9	52.8	49.2	44.6
	5%	WB	40.4	41.0	40.4	43.3	47.6	51.3	57.3	56.9	54.0	49.2	45.4	41.7
		MCDB	42.1	42.3	41.8	45.2	50.6	53.6	60.9	59.4	56.0	51.6	47.4	43.2
	10%	WB	38.6	39.5	39.0	42.2	46.2	50.0	55.7	55.7	52.9	47.9	43.6	39.9
		MCDB	40.6	41.2	40.9	44.1	48.5	52.4	58.5	57.9	55.0	50.2	45.7	41.9
Mean Daily Temperature Range	5% DB	MDBR	5.2	5.1	5.4	5.6	6.0	5.3	6.1	5.9	5.3	4.8	4.9	5.1
		MCDBR	6.1	5.3	5.5	6.4	9.0	8.2	10.4	7.9	6.3	4.9	4.8	6.1
		MCWBR	6.7	5.7	5.5	4.7	5.6	5.3	6.3	5.3	4.4	4.3	5.2	6.7
	5% WB	MCDBR	6.1	5.3	5.3	6.0	8.3	7.6	9.6	7.4	6.0	4.5	4.7	6.6
		MCWBR	6.9	5.8	5.7	4.9	5.6	5.4	6.2	5.3	4.7	4.6	5.7	7.5
Clear-Sky Solar Irradiance	taub	0.372	0.314	0.337	0.351	0.351	0.348	0.353	0.346	0.334	0.309	0.354	N/A	
	taud	2.437	2.157	2.281	2.376	2.452	2.502	2.509	2.521	2.498	2.256	2.433	N/A	
	Ebn at Noon	35	152	221	254	268	272	266	256	228	153	32	0	
	Edh at Noon	5	16	24	29	29	29	28	25	20	14	4	0	
All-Sky Solar Radiation	RadAvg	11	108	451	1003	1504	1670	1486	1102	599	200	27	0	
	RadStd	1	14	49	91	103	221	141	93	49	36	3	0	

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47)	Regional (15 neighbors)	+0.81	N/A	N/A	N/A	N/A	N/A	N/A	-221	N/A	N/A	

Nomenclature: See separate page